

QUANTUM COMPUTING AND ITS IMPLICATIONS WITH CYBERSECURITY

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BACHELOR OF COMPUTER APPLICATION

BY

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DEPARTMENT OF COMPUTER APPLICATIONS

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**KRISTU JYOTI COLLEGE OF MANAGEMENT AND TECHNOLOGY,
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CERTIFICATE

Certify that the seminar report entitled **“QUANTUM COMPUTING AND ITS IMPLICATIONS WITH CYBERSECURITY”** is a bonafide report of the seminar done by **ANAND S (230021080179)** under our guidance and supervision is submitted in partial fulfillment of the Bachelor of Computer Applications, awarded by Mahatma Gandhi University, Kerala and that no part of this work has been submitted earlier for the award of any other degree.

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DECLARATION

I hereby declare that the seminar work entitled “**Quantum Computing And Its Implications With Cybersecurity**” submitted to Mahatma Gandhi University in partial fulfillment of requirement for the award of degree of Bachelor of Computer Application from Kristu Jyoti College of Management and Technology, Changanacherry is a record of bonafide work done under the guidance of Ms.Aby Rose Varghese , seminar guide, Kristu Jyoti of Management and Technology,Changanacherry.This project has not been submitted in partial or fulfillment of any other degree/diploma/fellowship or similar of this university or any other university.

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ABSTRACT

Quantum computing is an emerging and rapidly advancing field of computing that is based on the principles of quantum mechanics such as superposition, entanglement, and quantum interference. Unlike classical computers, which process information using bits that exist in either a 0 or a 1 state, quantum computers use qubits that can exist in multiple states simultaneously. This unique capability enables quantum computers to perform certain complex computations at speeds far beyond the capabilities of classical systems. As a result, quantum computing has the potential to bring revolutionary changes in areas such as artificial intelligence, scientific simulations, optimization problems, and data analysis.

However, the same computational power that makes quantum computing highly promising also poses significant challenges to cyber security. Modern cyber security systems rely heavily on cryptographic algorithms to ensure secure communication, data confidentiality, authentication, and integrity. Most widely used encryption techniques, including RSA, Diffie–Hellman, and Elliptic Curve Cryptography, are based on mathematical problems that are difficult for classical computers to solve but can potentially be broken by quantum algorithms such as Shor’s algorithm. This creates a serious security concern, as sensitive data encrypted using current methods may become vulnerable in the future.

This seminar presents a detailed study of quantum computing and examines its implications on existing cyber security mechanisms. It discusses the limitations of current cryptographic systems in the presence of quantum computing and highlights emerging solutions such as post-quantum cryptography and quantum key distribution, which aim to provide secure communication in the quantum era. The seminar emphasizes the importance of early awareness, research, and migration toward quantum-resistant security technologies to protect critical digital information and ensure long-term cyber security.

Keywords: Quantum Computing, Cyber Security, Post-Quantum Cryptography

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1. INTRODUCTION

In the present digital era, cyber security has become one of the most critical concerns due to the rapid growth of internet-based services, cloud computing, online banking, e-commerce, and digital communication. Large volumes of sensitive data such as personal information, financial records, healthcare data, and national security information are transmitted and stored electronically. Protecting this data from unauthorized access, cyber attacks, and data breaches is a major challenge.

Traditional cyber security mechanisms rely heavily on cryptographic algorithms that are designed based on the limitations of classical computers. These algorithms assume that certain mathematical problems, such as factoring large numbers or solving discrete logarithms, are practically impossible to solve within a reasonable time using classical computing techniques. As a result, encryption methods like RSA and Elliptic Curve Cryptography have been widely adopted to secure digital communication.

Quantum computing introduces a new computing paradigm that challenges these assumptions. By utilizing principles of quantum mechanics, quantum computers can perform certain calculations exponentially faster than classical

computers. While this advancement opens new opportunities in science and technology, it also poses a serious threat to existing cyber security systems. This seminar aims to study the fundamentals of quantum computing, analyze its impact on current cyber security practices, and explore possible solutions to ensure secure communication in the quantum era.

2. LITERATURE REVIEW

Extensive research has been carried out to examine the impact of quantum computing on cyber security, particularly in the field of cryptography. Early studies in cyber security focused on developing encryption algorithms that could protect sensitive information from attacks by classical computers. Public-key cryptographic systems such as RSA, Diffie–Hellman, and Elliptic Curve Cryptography became the foundation of secure digital communication due to their reliance on complex mathematical problems that are difficult for classical computers to solve.

A major breakthrough in this area was achieved by **Peter W. Shor**, who introduced **Shor's algorithm**, a quantum algorithm capable of efficiently factoring large integers and solving discrete logarithm problems. This discovery

demonstrated that cryptographic systems considered secure against classical attacks could be compromised by sufficiently powerful quantum computers. As a result, widely used public-key encryption techniques such as RSA and Diffie–Hellman were identified as vulnerable in the quantum computing era.

Another significant contribution to quantum cryptanalysis was **Grover’s algorithm**, which showed that quantum computers could search unsorted databases faster than classical systems. Although Grover’s algorithm does not completely break symmetric encryption algorithms such as AES, it effectively reduces their security strength by enabling faster brute-force attacks. This finding indicates that larger key sizes are required to maintain an adequate level of security in the presence of quantum computing.

Recent research has shifted its focus toward developing **post-quantum cryptography**, which aims to design cryptographic algorithms that can resist both classical and quantum attacks. These algorithms are based on mathematical problems that are believed to be difficult even for quantum computers. Organizations such as the **National Institute of Standards and Technology (NIST)** are actively involved in evaluating

and standardizing quantum-resistant cryptographic algorithms to ensure future security.

In addition to post-quantum cryptography, researchers have explored **quantum key distribution (QKD)** as a secure method for exchanging cryptographic keys. QKD uses the principles of quantum mechanics to detect any eavesdropping attempt during key exchange, providing a theoretically secure communication channel. Overall, existing literature highlights the serious threat posed by quantum computing to current cyber security systems and emphasizes the urgent need to transition toward quantum-safe security solutions.

3. METHODOLOGY / SYSTEM DESCRIPTION

Quantum computing is fundamentally different from classical computing in terms of how information is represented and processed. Classical computers use **bits** as the basic unit of data, where each bit can exist only in one of two states: 0 or 1. All computations in classical systems are performed by manipulating these bits using logical operations. Although classical computers are highly efficient for many applications, they face limitations when solving complex problems that require massive parallel processing.

In contrast, quantum computers use **quantum bits or qubits** as the basic unit of information. A qubit can exist in a combination of both 0 and 1 states simultaneously due to a quantum mechanical property known as **superposition**. This allows quantum computers to process a large number of possibilities at the same time. In addition to superposition, qubits can also exhibit **entanglement**, a phenomenon in which the state of one qubit is directly related to the state of another qubit, even when they are physically separated. Entanglement enables strong correlations between qubits and plays a crucial role in increasing the computational power of quantum systems.

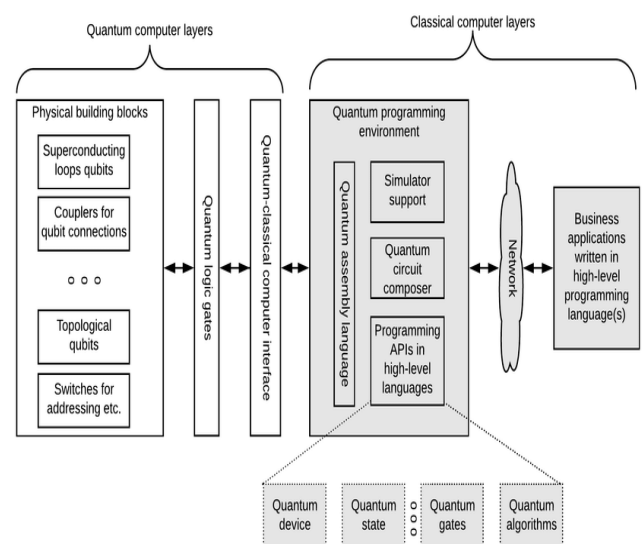
3.1 Working Principle of Quantum Computing

The working of a quantum computer involves the manipulation of qubits using **quantum gates**. Quantum gates are the quantum equivalent of logical gates in classical computers, but they operate on qubits and exploit quantum properties. These gates control the superposition and entanglement of qubits to perform computations.

Quantum algorithms are specifically designed to take advantage of quantum

principles such as superposition, entanglement, and **quantum interference**. Interference is used to amplify correct computational results while suppressing incorrect ones. Through a sequence of quantum gate operations, a quantum algorithm processes information in a highly parallel manner.

At the end of computation, the quantum state of the qubits is **measured**, which collapses the superposition into a definite classical output. This output can then be interpreted using classical computing systems. Due to this approach, quantum computers can solve certain mathematical and computational problems much faster than classical computers, especially problems related to factorization, optimization, and cryptography.



3.2 Impact on Cyber Security Systems

Modern cyber security systems rely heavily on **public-key cryptography** to ensure secure key exchange, authentication, and data protection. Encryption algorithms such as RSA, Diffie–Hellman, and Elliptic Curve Cryptography are based on mathematical problems that are extremely difficult for classical computers to solve within a practical time frame.

However, quantum computing introduces serious challenges to these systems. Quantum algorithms, particularly **Shor’s algorithm**, can efficiently solve the mathematical problems on which public-key cryptography is based. This means that encryption methods currently considered secure may become vulnerable once sufficiently powerful quantum computers are available. As a result, sensitive data transmitted over networks, including financial transactions, confidential communications, and personal information, could be exposed to security risks.

3.3 Proposed Approaches to Mitigate Quantum Threats

To address the security challenges introduced by quantum computing,

researchers and security experts have proposed two major approaches:

1. **Post-Quantum Cryptography:**

This approach focuses on developing new cryptographic algorithms that can run on classical computers but are resistant to both classical and quantum attacks. These algorithms are based on mathematical problems that are believed to be difficult even for quantum computers to solve. Post-quantum cryptography aims to replace or upgrade existing encryption systems without requiring quantum hardware.

2. **Quantum-Based Security Mechanisms:**

This approach uses the principles of quantum mechanics to enhance security. One prominent example is **quantum key distribution (QKD)**, which allows secure exchange of cryptographic keys by detecting any eavesdropping attempt during transmission. QKD provides a high level of security based on the laws of physics rather than computational complexity.

Together, these approaches aim to ensure secure communication and data protection in the emerging quantum computing era.

4. FACTS AND FINDINGS

The study of quantum computing and its implications on cyber security reveals several important facts and observations that highlight the urgency of preparing for the quantum era. These findings emphasize both the risks posed by quantum computing and the need for adopting quantum-safe security mechanisms.

One of the most significant findings is that **quantum computers have the potential to completely break widely used public-key encryption algorithms** such as RSA, Diffie–Hellman, and Elliptic Curve Cryptography. These encryption techniques form the backbone of secure communication on the internet, including online banking, secure websites, and digital authentication systems. Quantum algorithms like Shor’s algorithm can efficiently solve the mathematical problems on which these encryption methods are based, making them vulnerable in the presence of powerful quantum computers.

Another important observation is that **symmetric encryption algorithms such as AES are relatively more secure against quantum attacks** when compared to public-key encryption. However, quantum algorithms like Grover’s algorithm reduce their effective security

strength. As a result, larger key sizes are required to maintain an adequate level of protection. This implies that existing systems using symmetric encryption must be upgraded to stronger configurations to remain secure in the quantum era.

The study also highlights the emergence of a new category of cyber attacks known as **“harvest now, decrypt later.”** In this approach, attackers collect and store encrypted data today with the intention of decrypting it in the future using quantum computers. This poses a serious threat to long-term confidential information such as government records, medical data, and intellectual property that must remain secure for many years.

Another key finding is that **quantum key distribution (QKD)** provides a highly secure method for cryptographic key exchange. QKD uses the principles of quantum mechanics to detect any eavesdropping attempt during communication, thereby ensuring the integrity and confidentiality of the exchanged keys. This makes QKD a promising solution for secure communication in the quantum era.

Finally, the study finds that **the transition to quantum-safe cyber security systems is a complex and time-consuming process.** Migrating existing infrastructure

to quantum-resistant algorithms requires careful planning, testing, and standardization. Therefore, organizations must initiate this transition well in advance, even before large-scale quantum computers become widely available, to ensure long-term data security.

5. RESULTS AND DISCUSSION

- The analysis of quantum computing and its implications on cyber security reveals both positive and negative outcomes. On one hand, quantum computing introduces powerful computational capabilities that challenge existing security mechanisms. On the other hand, it drives significant advancements in the development of stronger and more secure cyber security solutions. This section discusses the major advantages and limitations identified through the study.

5.1 Advantages

- One of the key results of this study is that the emergence of quantum computing encourages the development of stronger and more advanced security mechanisms. As traditional cryptographic systems become vulnerable to quantum

attacks, researchers and security experts are motivated to design new encryption techniques that can withstand both classical and quantum threats.

- Quantum computing also drives innovation in cryptographic research and secure communication technologies. The need to protect data in the quantum era has led to increased research in areas such as post-quantum cryptography and quantum-based security solutions. These advancements contribute to the overall improvement of cyber security frameworks.
- Another important advantage is the enablement of secure communication methods such as **quantum key distribution (QKD)**. QKD uses the principles of quantum mechanics to securely exchange cryptographic keys and detect any unauthorized interception. This provides a higher level of security compared to traditional key exchange mechanisms.
- Furthermore, quantum-safe security approaches promote long-term protection of sensitive data against future threats. By adopting quantum-resistant encryption techniques, organizations can

ensure that confidential information such as financial records, medical data, and government communications remains secure even as quantum computing technology advances.

5.2 Limitations

- Despite its advantages, quantum computing also presents several limitations and challenges in the context of cyber security. One major limitation is that practical quantum computers are expensive and complex to build, operate, and maintain. The technology is still in its early stages and requires specialized hardware, controlled environments, and skilled expertise.
- Another limitation is that post-quantum cryptographic algorithms may require higher computational resources compared to traditional encryption techniques. This can result in increased processing time, memory usage, and power consumption, especially in resource-constrained systems.
- Transitioning existing systems to quantum-safe solutions is also a challenging and time-consuming process. Many current digital

infrastructures are deeply integrated with traditional cryptographic algorithms, making migration complex and costly. Organizations must carefully plan and implement this transition to avoid security gaps.

- Additionally, there is a lack of awareness and expertise in quantum security technologies among professionals and organizations. The limited availability of trained experts and practical knowledge can slow down the adoption of quantum-safe cyber security solutions.

6. CONCLUSION AND FUTURE SCOPE

Quantum computing represents a significant technological breakthrough that has the potential to transform various fields of computing, including cyber security. By utilizing the principles of quantum mechanics such as superposition and entanglement, quantum computers are capable of solving complex problems much faster than classical computers. While this advanced computational power opens new opportunities in scientific research and technology, it also poses

serious challenges to existing cyber security systems.

The seminar concludes that current cryptographic mechanisms, particularly public-key encryption techniques, are vulnerable to quantum attacks. As quantum computers continue to develop, encryption methods that are secure today may become ineffective in the future. This highlights the urgent need for organizations, governments, and security professionals to prepare for the quantum era. Adopting **quantum-resistant cryptographic algorithms** and secure communication techniques is essential to protect sensitive data and maintain trust in digital systems.

Looking toward the future, research is expected to focus on developing **scalable, reliable, and cost-effective quantum computers** that can be used in practical applications. At the same time, significant efforts are being made to improve and standardize **post-quantum cryptographic algorithms** that can provide long-term security against both classical and quantum attacks. Integration of quantum security solutions such as **quantum key distribution** into real-world communication networks is also an important area of future development.

As quantum technologies continue to evolve, cyber security strategies must also advance to address new threats and challenges. Ensuring the **confidentiality, integrity, and availability** of digital information in the quantum era requires continuous research, awareness, and proactive adoption of quantum-safe security solutions.

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